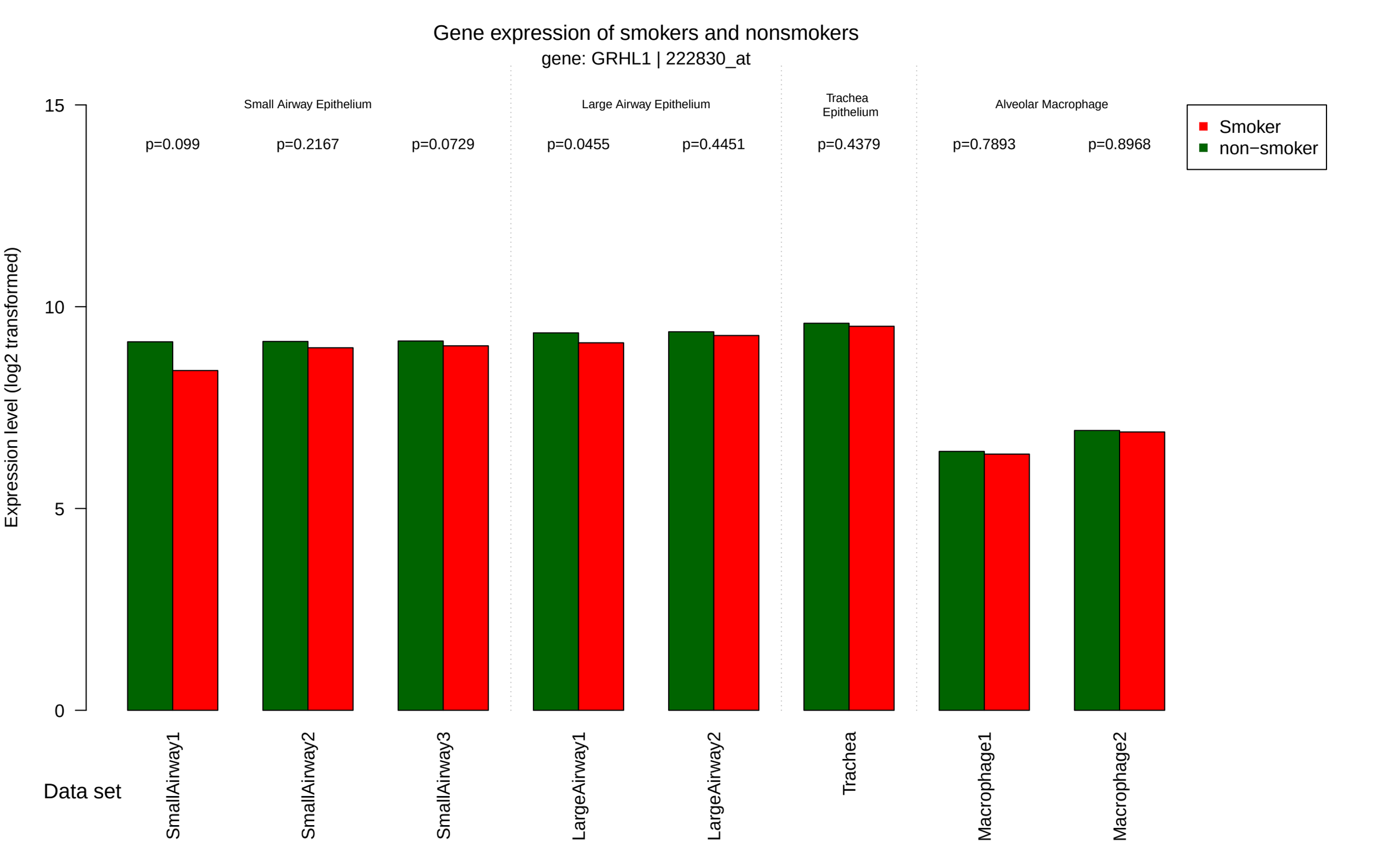
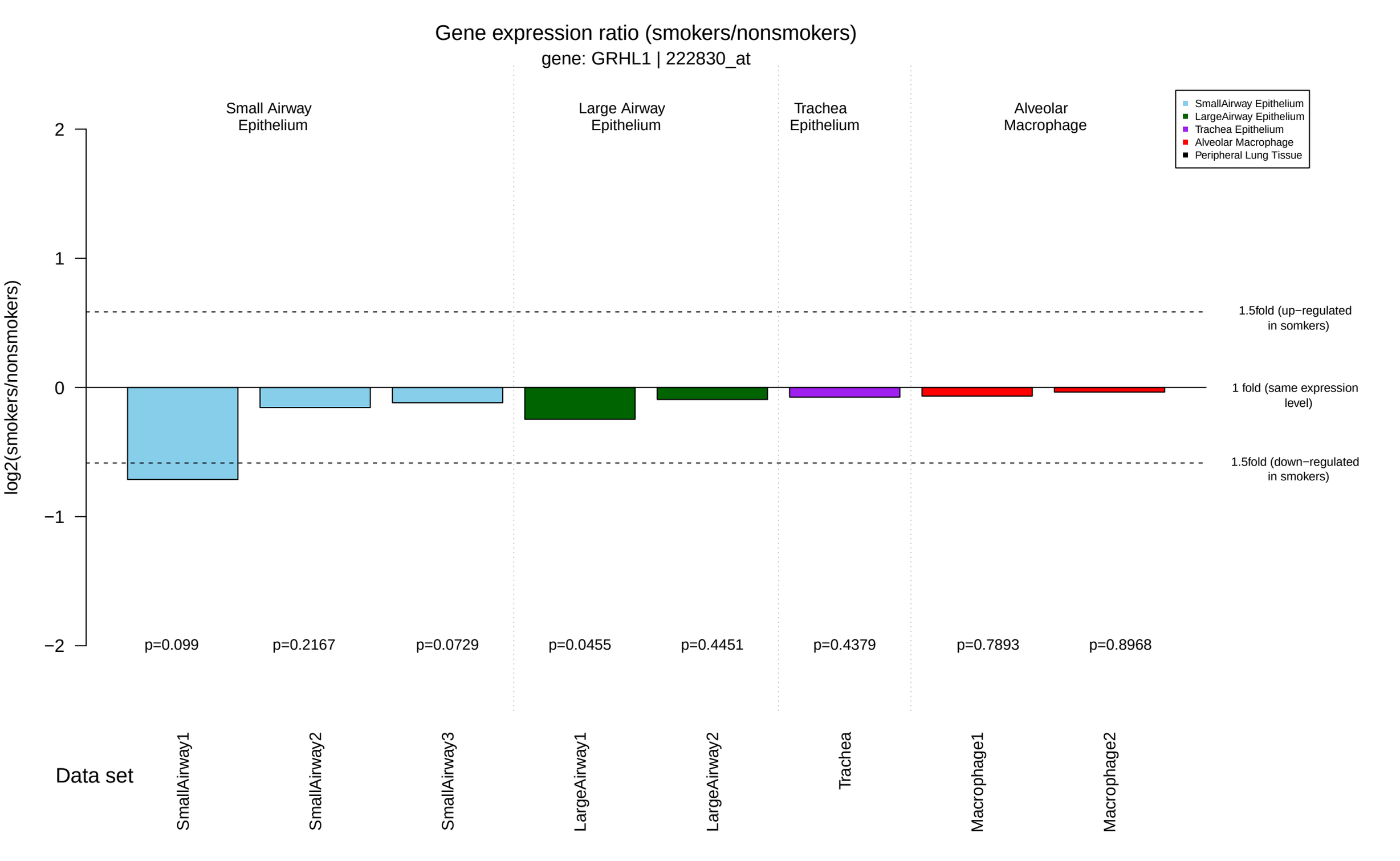
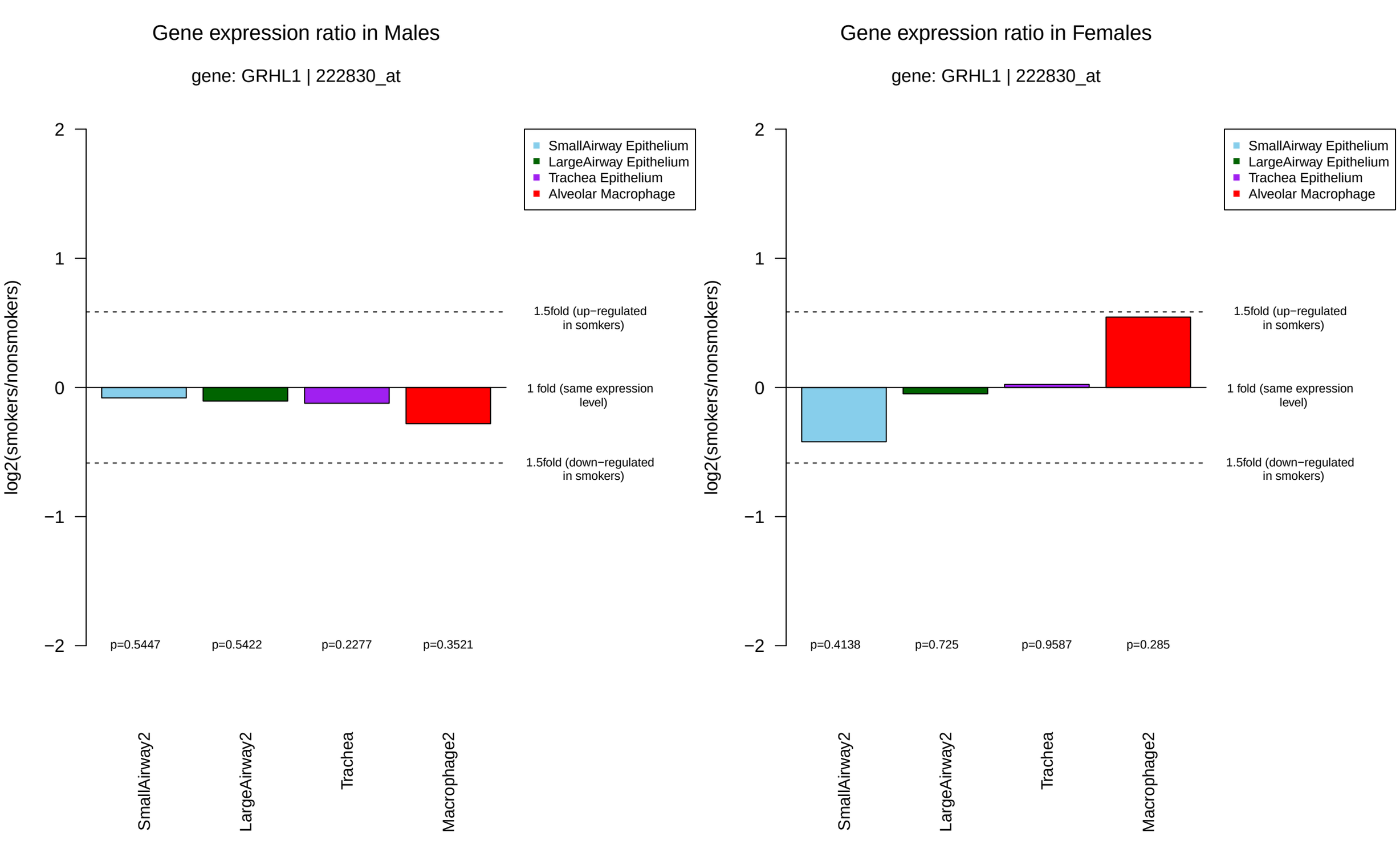
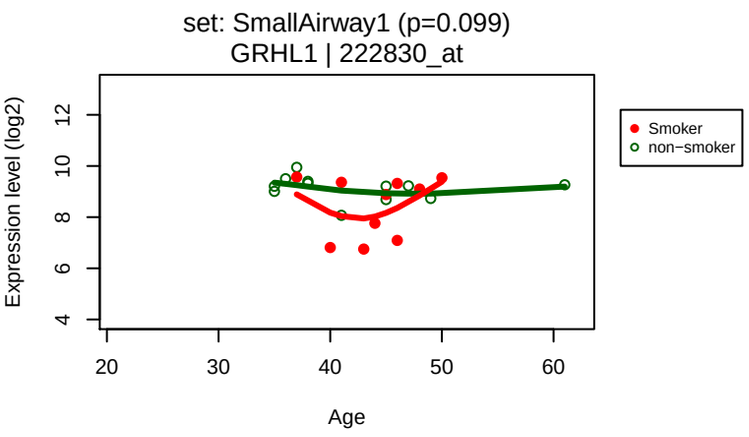
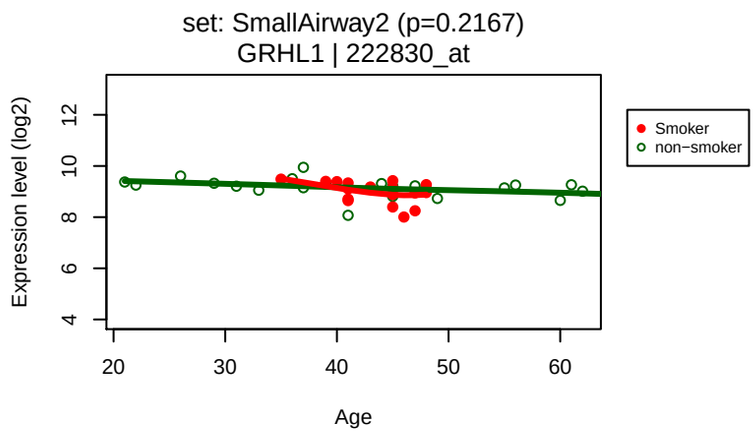
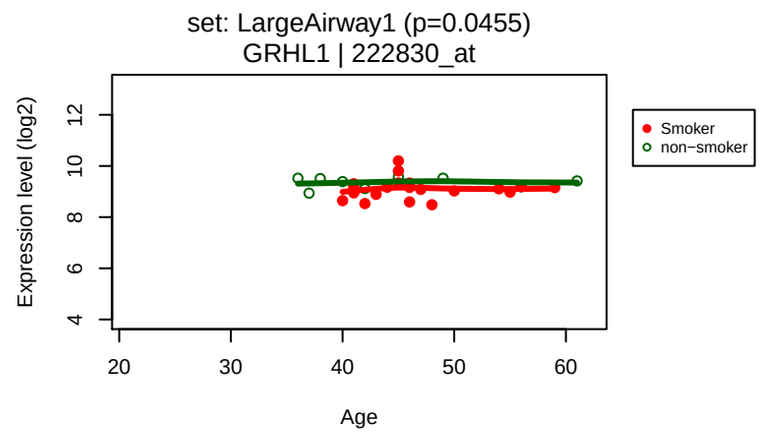

Gene expression of smokers and nonsmokers
gene: GRHL1 | 222830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.099
p=0.2167
p=0.0729
p=0.0455
p=0.4451
p=0.4379
p=0.7893
p=0.8968
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

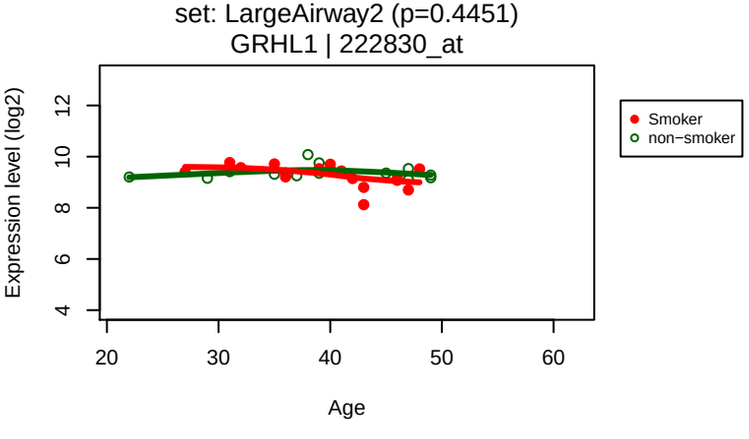

Gene expression ratio (smokers/nonsmokers)
gene: GRHL1 | 222830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.099
p=0.2167
p=0.0729
p=0.0455
p=0.4451
p=0.4379
p=0.7893
p=0.8968
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

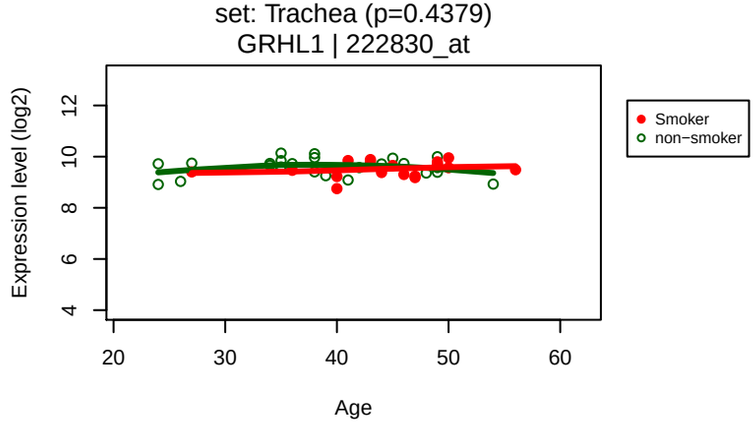

Gene expression ratio in Males
gene: GRHL1 | 222830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5447
p=0.5422
p=0.2277
p=0.3521
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRHL1 | 222830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4138
p=0.725
p=0.9587
p=0.285
SmallAirway2
LargeAirway2
Trachea
Macrophage2

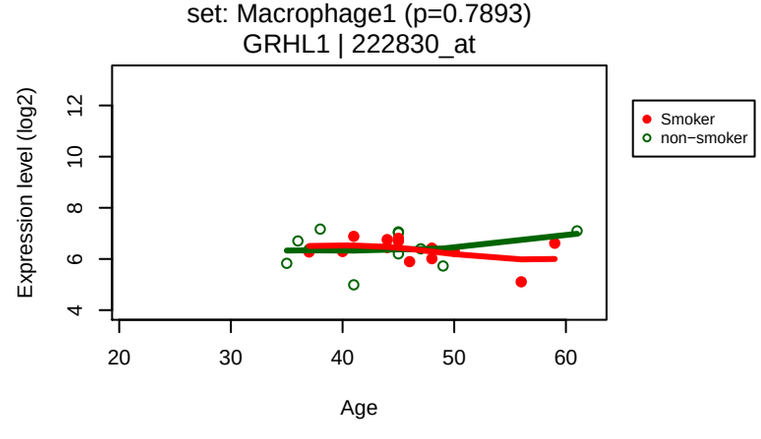

set: SmallAirway1 (p=0.099)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker

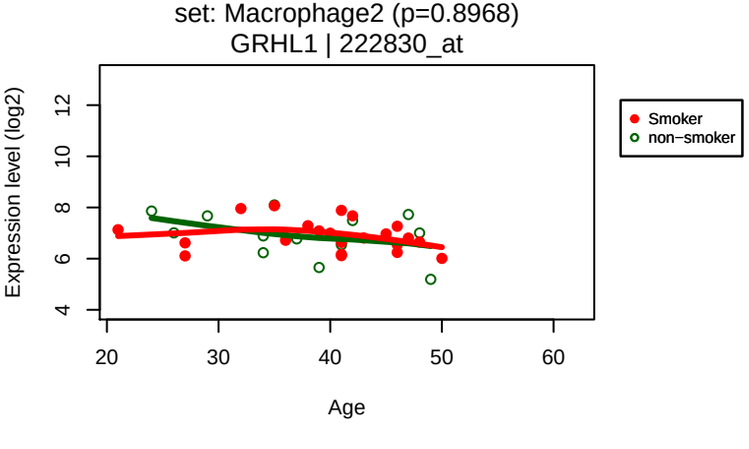

set: SmallAirway2 (p=0.2167)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker

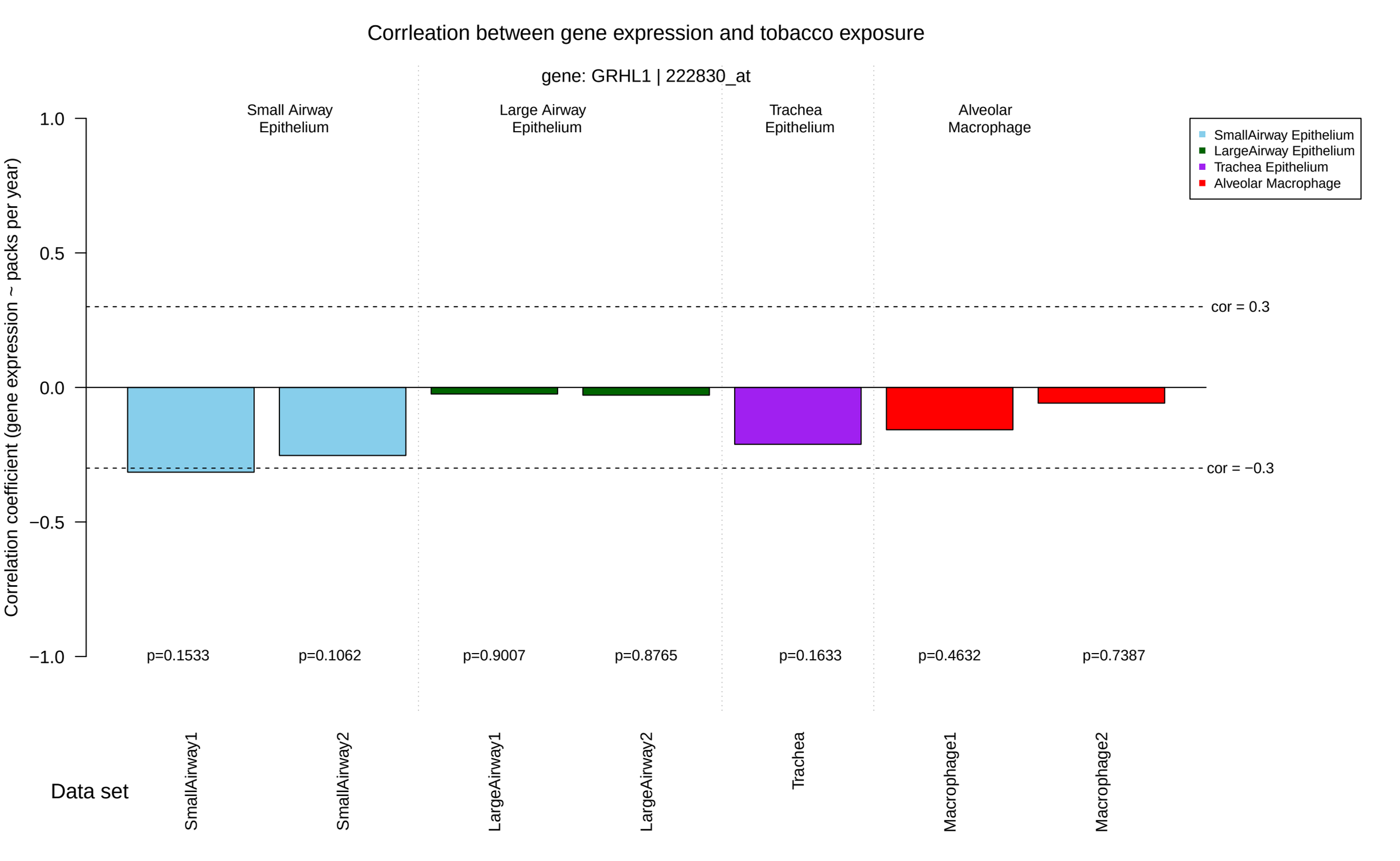

set: LargeAirway1 (p=0.0455)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4451)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4379)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7893)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8968)
GRHL1 | 222830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRHL1 | 222830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1533
p=0.1062
p=0.9007
p=0.8765
p=0.1633
p=0.4632
p=0.7387
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2